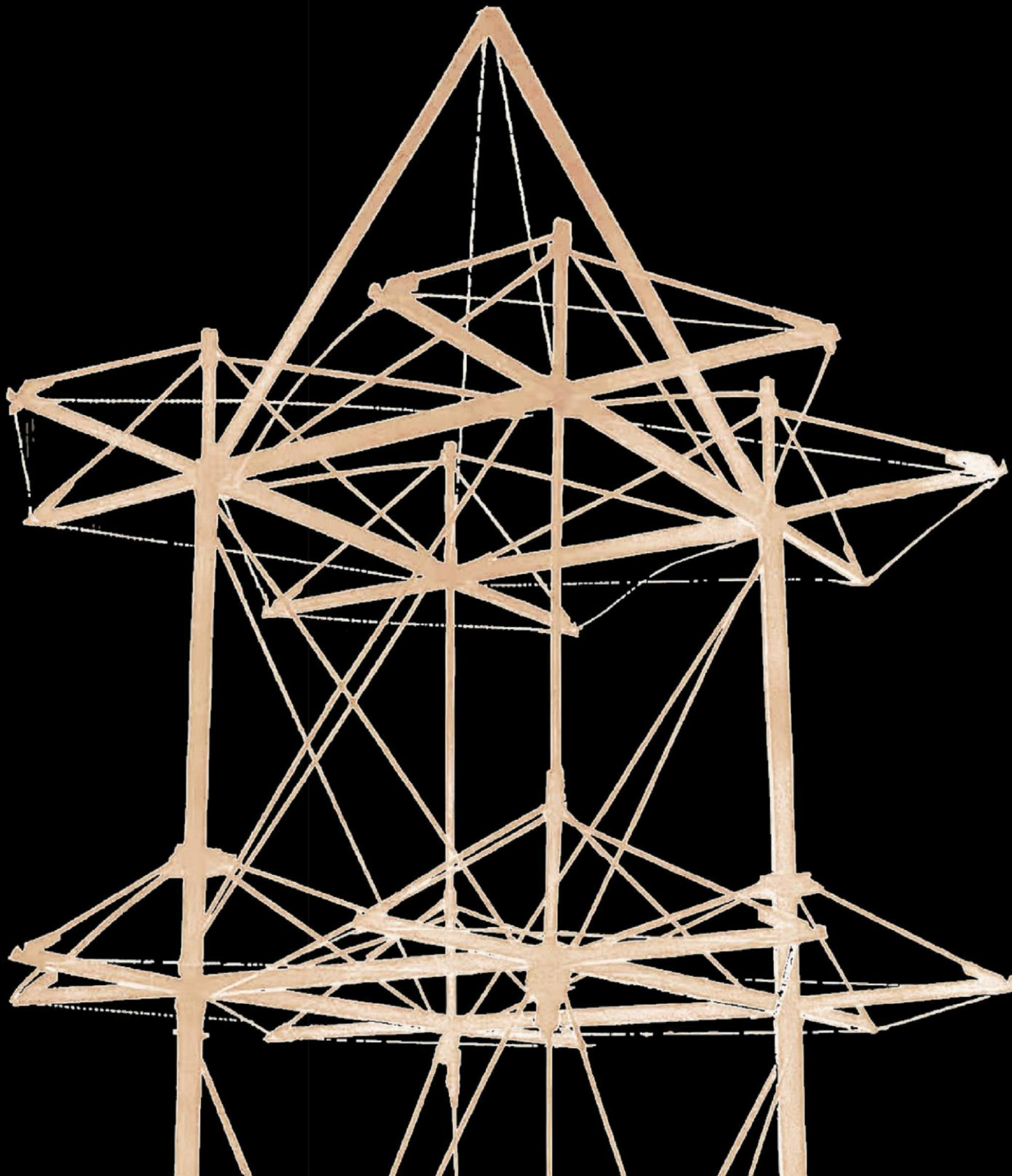
PORTFOLIO

BY SONG JINWEN



CONTENTS

"Innovation is not confined to the finished product,
but... in the moment where creativity and technology collide,
breaking the boundaries of what was once thought impossible."

01

AC Units as Urban

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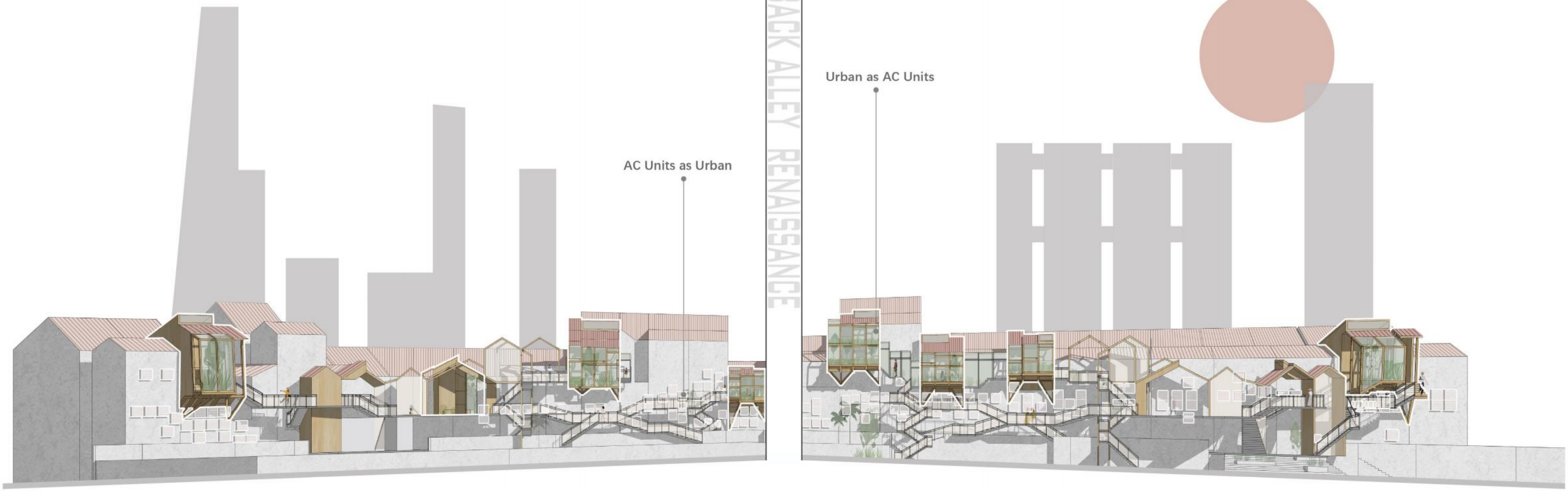
Urban as AC Units

BACK ALLEY RENAISSANCE



Design Descrption

The core of this design focuses on exploring and utilizing the urban voids as a typological format to seed new urban conditions. Within the highly built up fabric of Tanjong Pagar, Singapore, the design will concentrate on a pre-existing slippages and gaps in the urban fabric. The design inspiration stems from observing shophouse workers resting in the voids created by air conditioning units in the back alleys of shophouses and residents reclining on benches concealed by shrubbery. These scenarios have driven me to explore space form characteristics that attract people to linger. After the survey of resting spaces of different groups of people, it has shown that irregular planar and sectional boundaries can effectively reduce visual disturbances, providing a natural sense of spatial delineation, thereby enhancing the users' sense of safety and privacy.

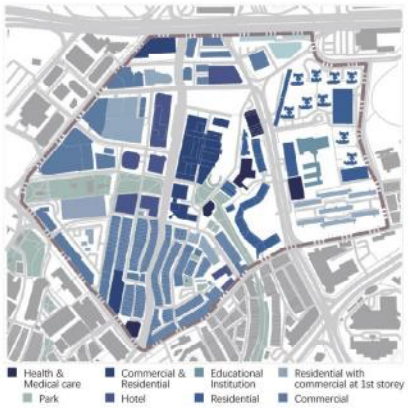
Based on this theory, the design adopts small concave and convex volumes as the basic units. These small volumes, through combination and variation, form functional elements such as tables, chairs, windows, etc.. By arranging these functional spaces—retail spaces, resting spaces, and circulation spaces—in a planar manner, spatial variations are created. Lastly, considering the distance between air conditioning units and buildings, different functional spaces exhibit various concave and convex sectional forms. These combinations and variations serve as a new architectural vocabulary, creating a series of rich and intriguing spatial forms. This design not only meets the needs of people to linger in various scenarios but also enhances the users' sense of place and belonging through changes in spatial forms.

Another design direction comes from the urban design level. Analysis of the current state of back alley of shophouses reveals that these areas are underutilized and lacking vitality. The further the distance from the main street, the less human activity there is. The design leverages the architectural vocabulary inspired by the site itself for redevelopment, ultimately identifying eight suitable locations for extensions. These eight units will revitalize the back alleys, transforming them into engaging public activity spaces.

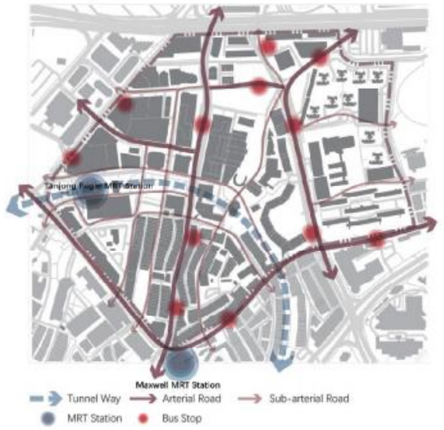
Macro Site Analysis

Site Model

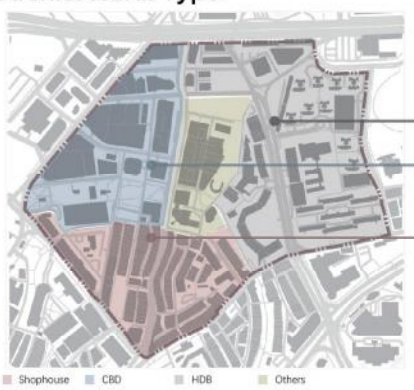
Land Use



Transportation

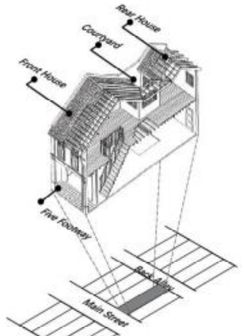


Architectural Type



Shophouse

A shophouse is a building type serving both as a commercial business at the first floor and a residence at the second floor or above. It is a building type found in Southeast Asia that is "a shop opening on to the pavement and also used as the owner's residence", and became a commonly used term since the 1950s.

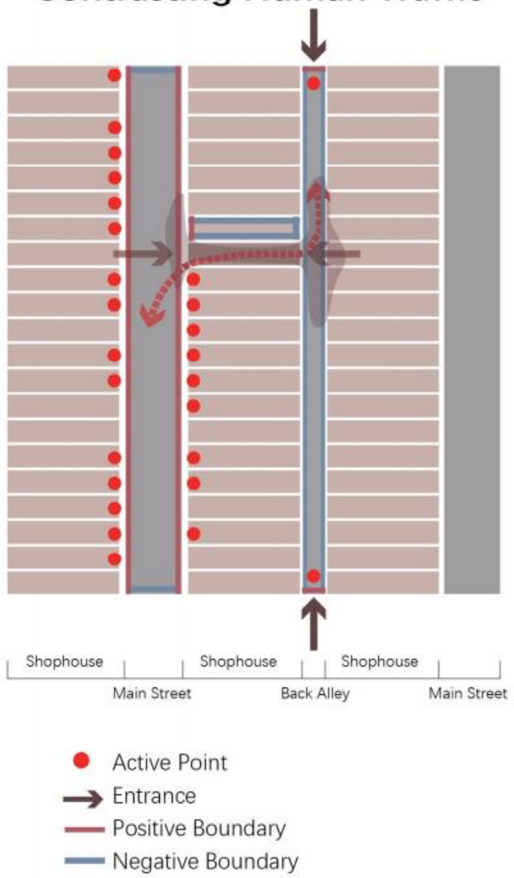


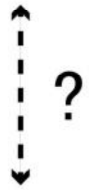
CBD

CBD (central business district) is the commercial and business center of a city. Guoco Tower is a mixed-use development skyscraper in Tanjong Pagar of the Downtown Core district of Singapore. With a height of 290m, it is currently the tallest building in Singapore.

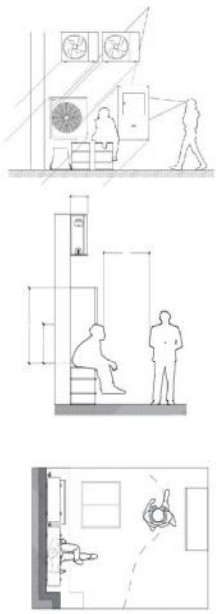
HDB

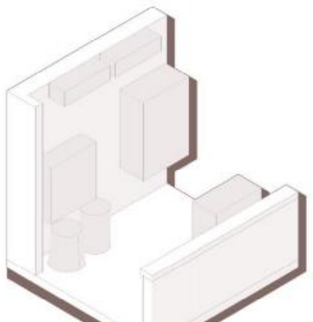
Contrasting Human Traffic





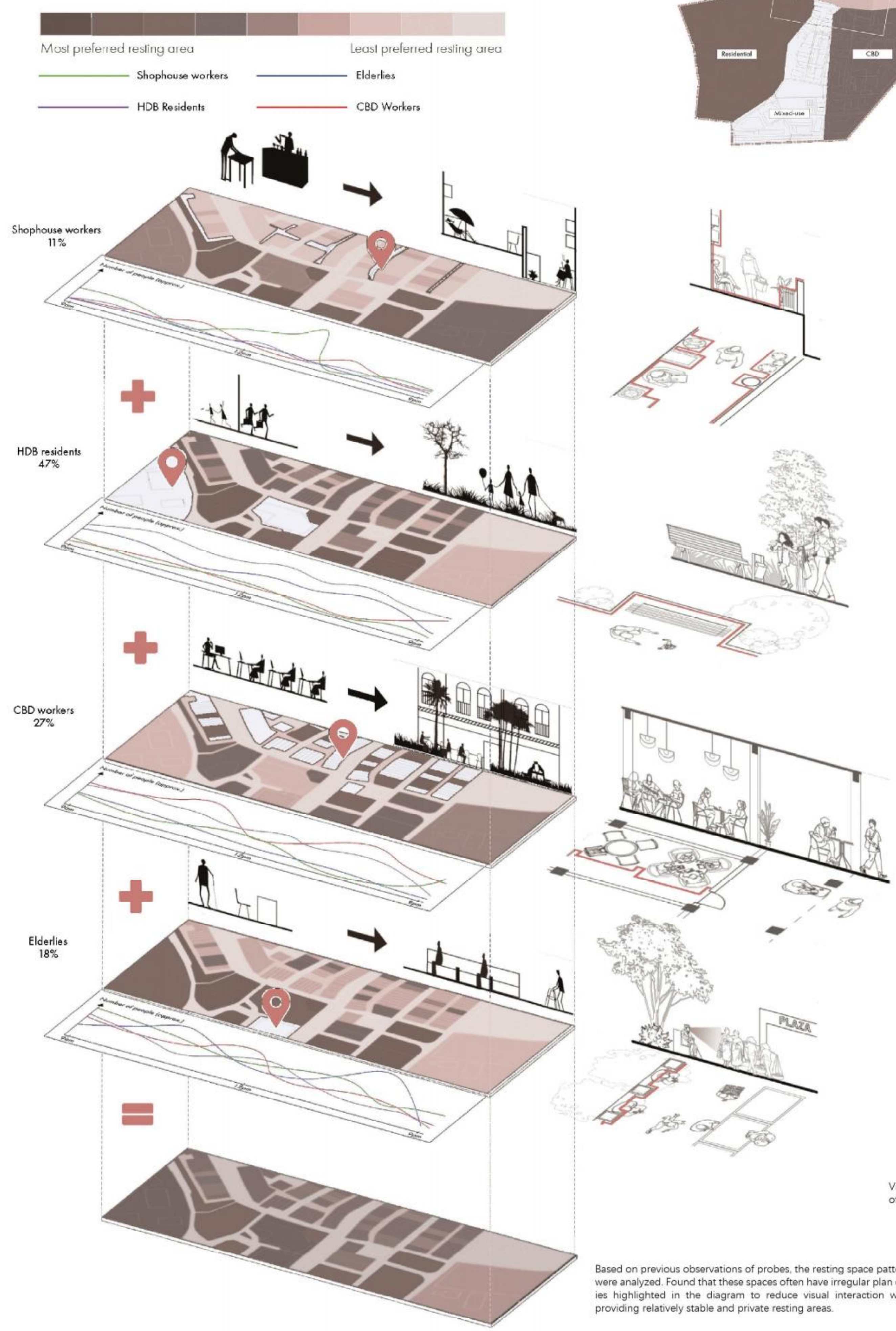
Probe Informal Resting Space

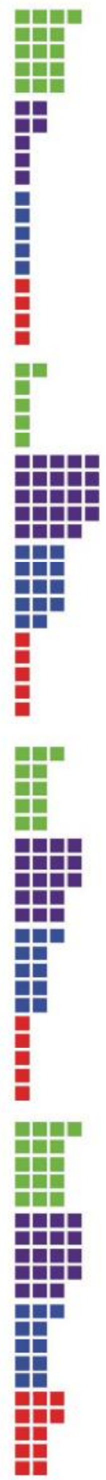




A shophouse worker is resting in the gap between air conditioning units in the back alley.

Analysis of Resting Spaces





Visualizaton of num. of people at 12P.M.

Based on previous observations of probes, the resting space patterns for four groups were analyzed. Found that these spaces often have irregular plan or section boundaries highlighted in the diagram to reduce visual interaction with others, thereby providing relatively stable and private resting areas.

Micro Site Selection

"d" refers to the distance from one point in the backalley to the nearst main street space.

- d<25m
- 25m<d<50m
- 50m<d<100m
- d>100m
- Where back alley meets main street
- Micro sites

Micro Site 1

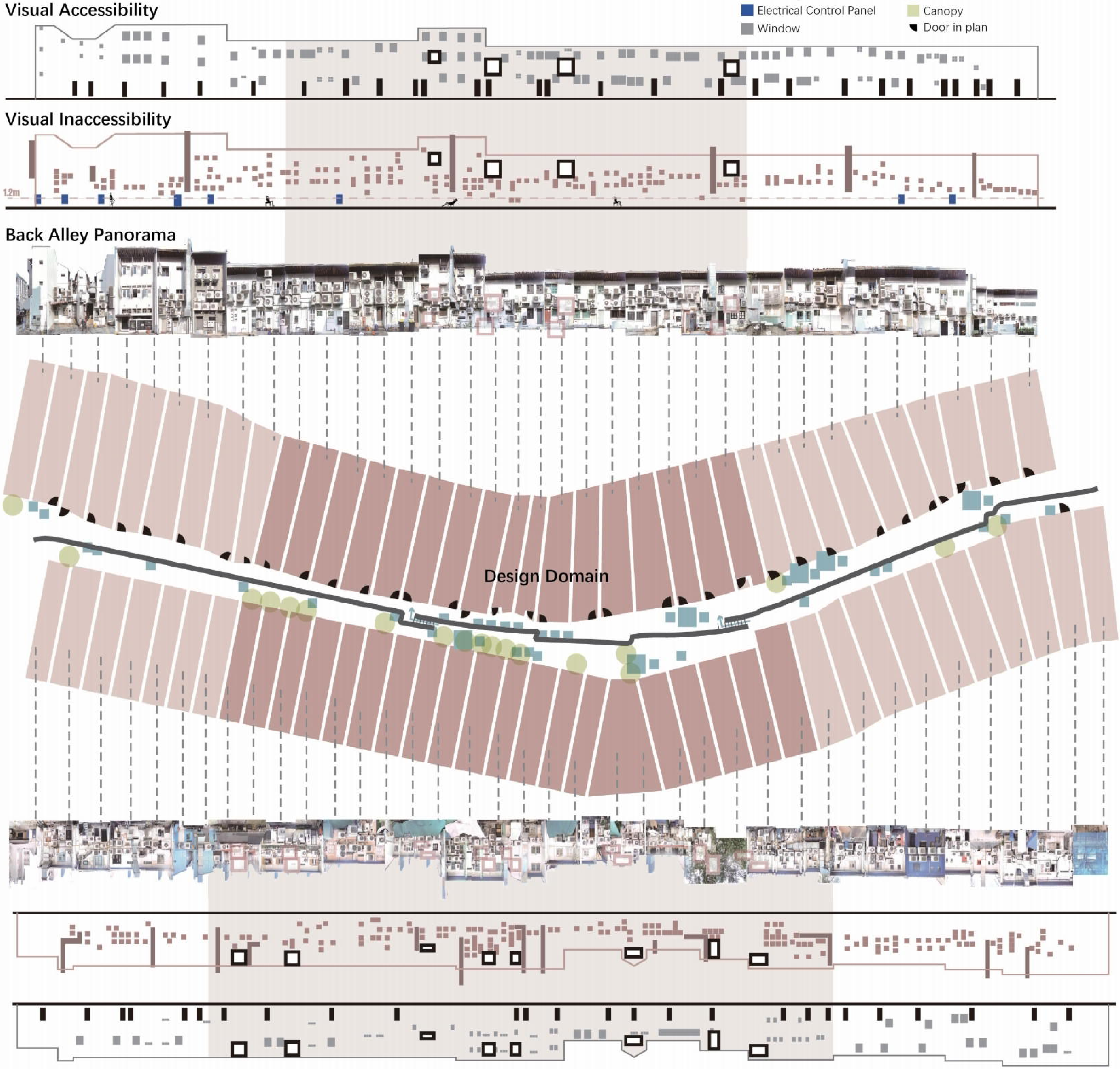
Micro Site 2

Micro Site 3

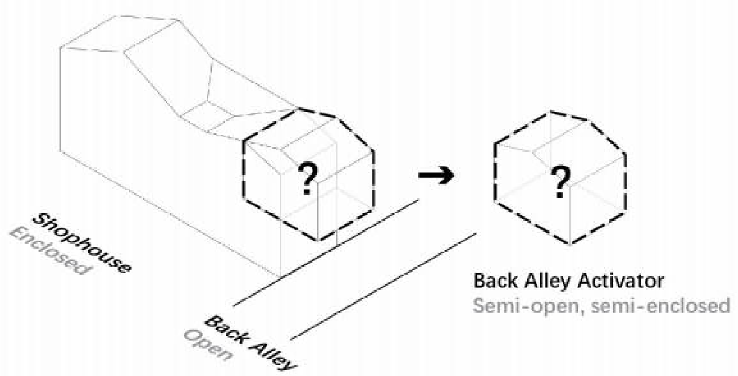
Based on preliminary analysis, spaces intended for people to rest should possess some certain spatial form characteristics. Whether it is an air conditioning unit in the back alley or small bushes on a lawn, they can both fulfill this role. Considering that most of the space in the back alley is underutilized and the fact that the farther away from the main street, the fewer the activities of people, the location is chosen in the deepest area of the back alley. The aim is to activate this area through architectural design, attracting more people to stay here. The design will make full use of the space in the back alley, creating a new space between the shophouse and the back alley, and utilizing the space formed by air conditioning units for the design.

Shophouse Back Alley Panorama Analysis

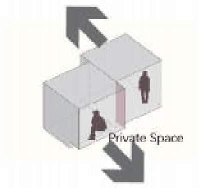
- Aircon Unit
- Chimney
- Electrical Control Panel
- Window
- Door in elevation
- Furniture
- Canopy
- Door in plan



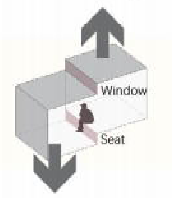
Design Strategy



Plan Logic

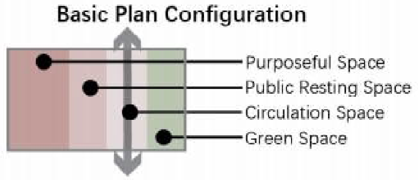


Section Logic

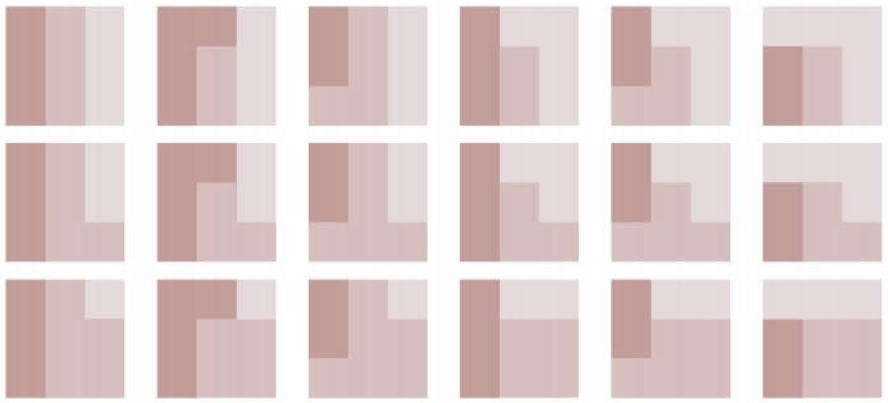


Plan Logic

Basic Plan Configuration

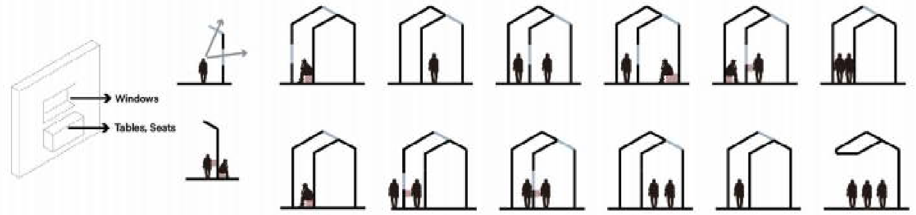


Permutation

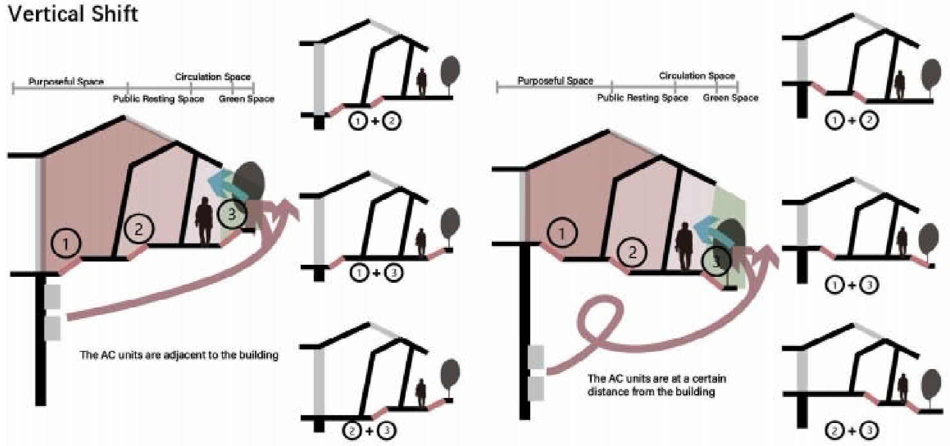


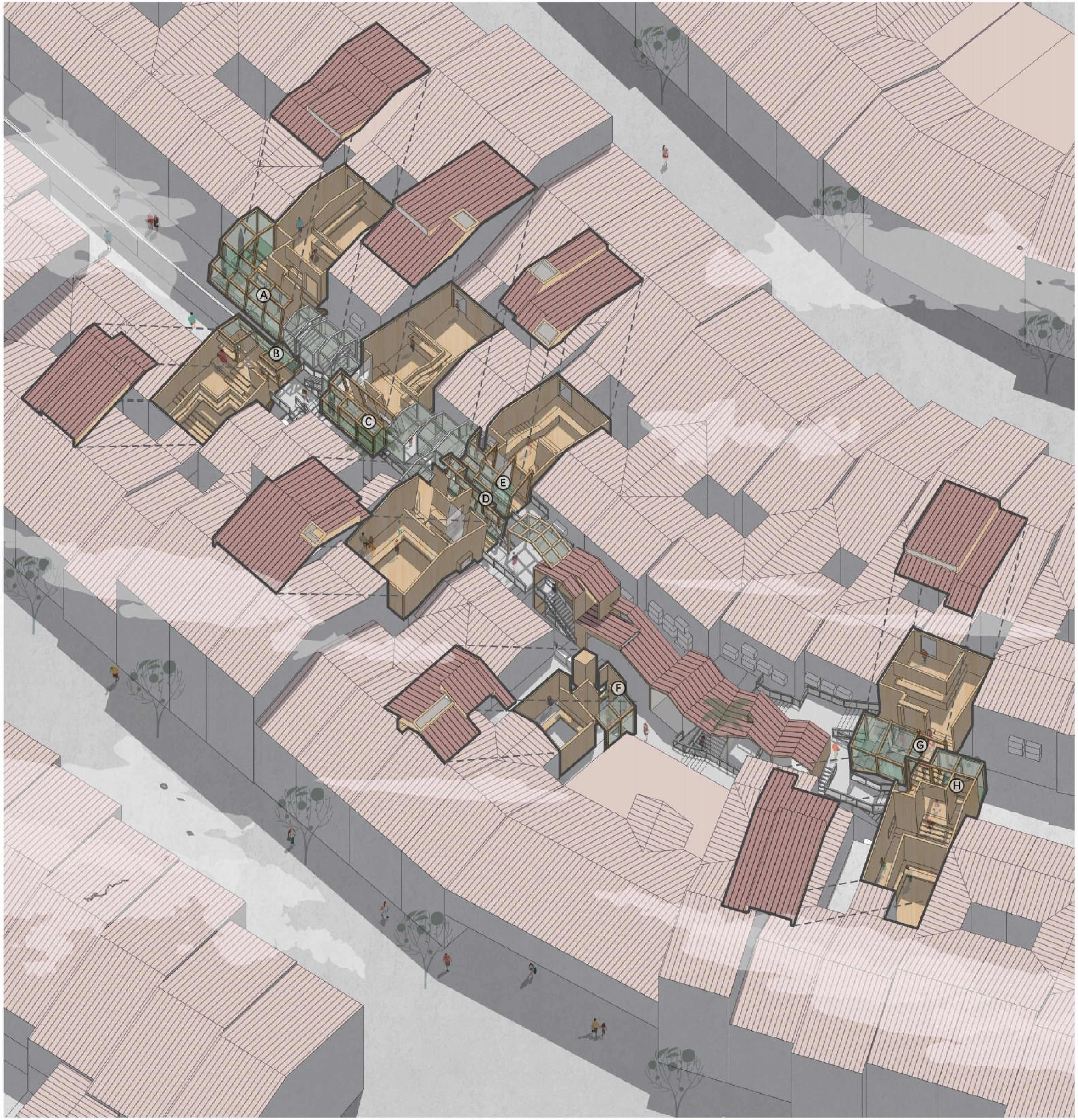
Section Logic

Horizontal Shift



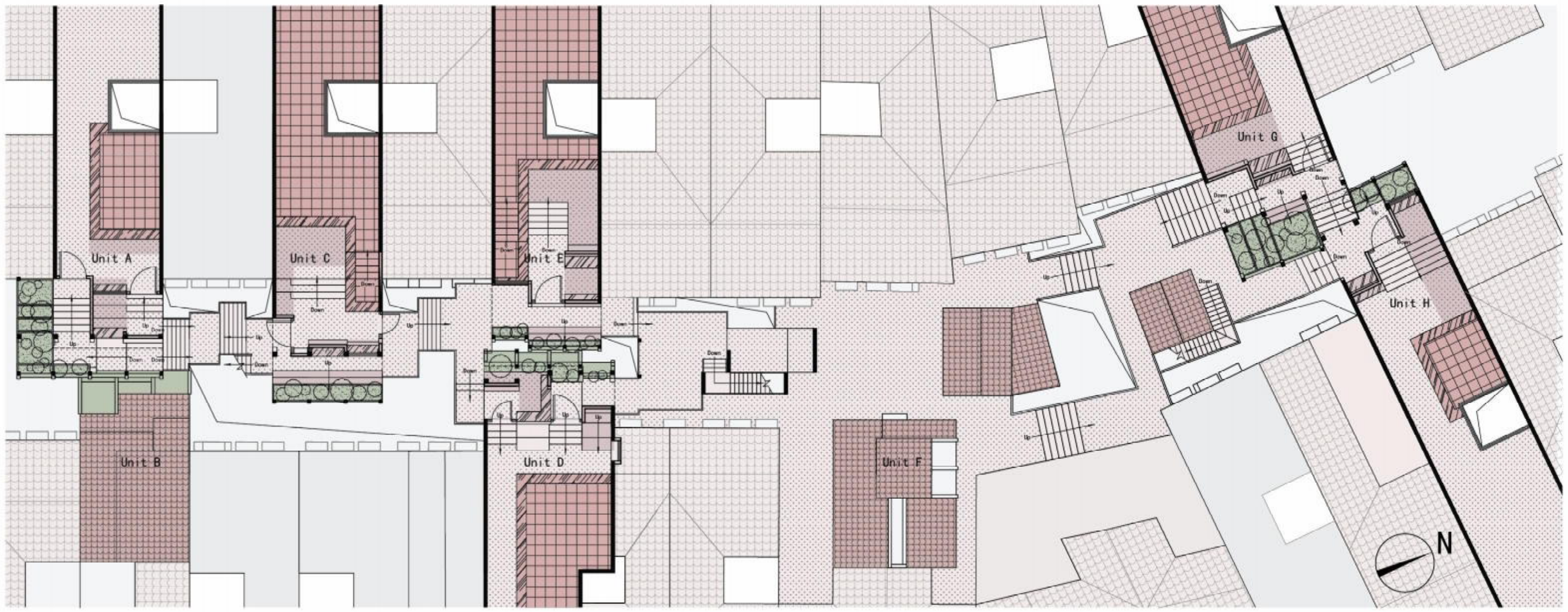
Vertical Shift





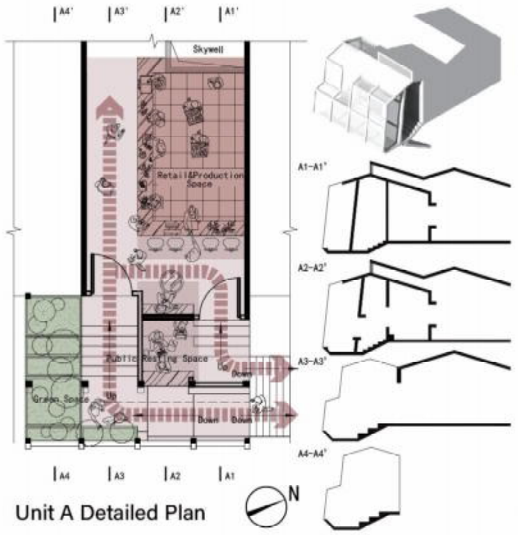
Bird's Eye View along the Back Alley

Overall Floor Plan

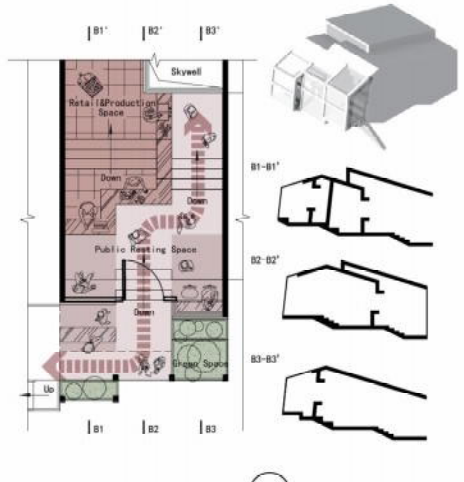


Unit Analysis

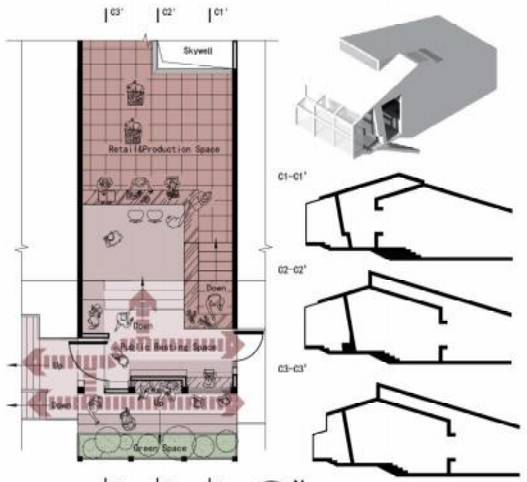
0 2 10 20m



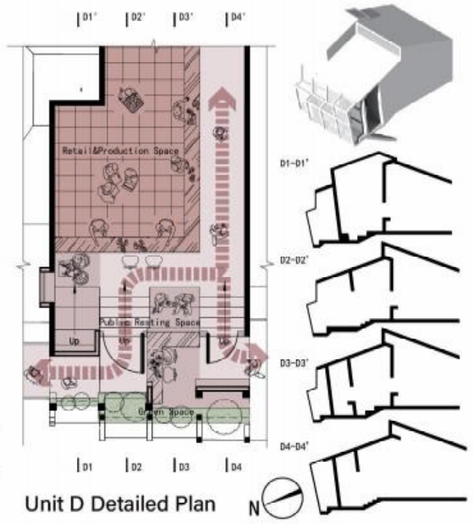
Unit A Detailed Plan



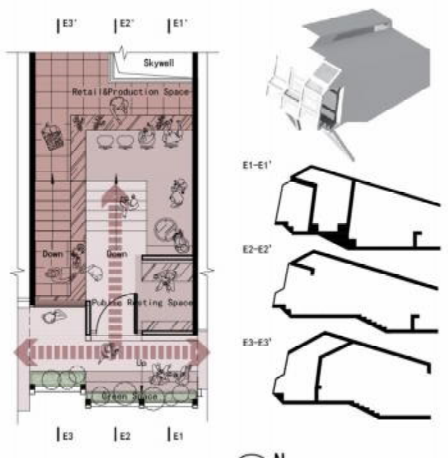
Unit B Detailed Plan



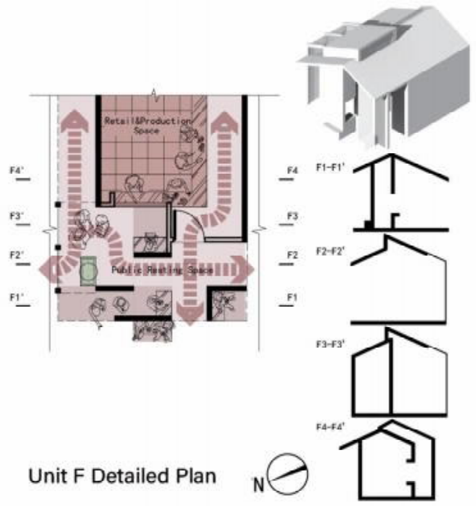
Unit C Detailed Plan



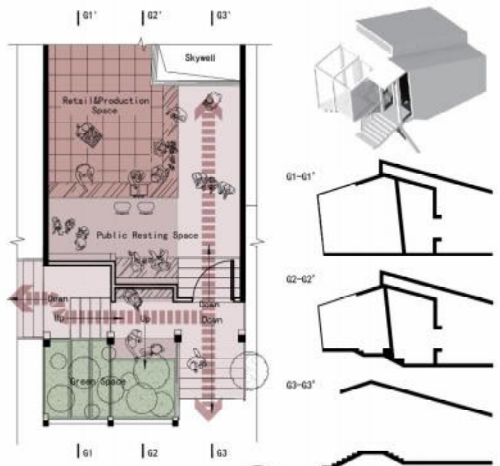
Unit D Detailed Plan



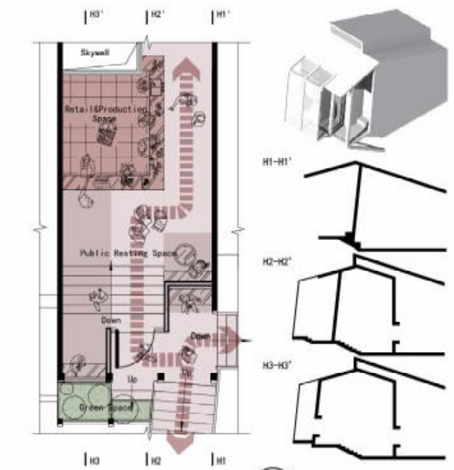
Unit E Detailed Plan



Unit F Detailed Plan



Unit G Detailed Plan



Unit H Detailed Plan

02

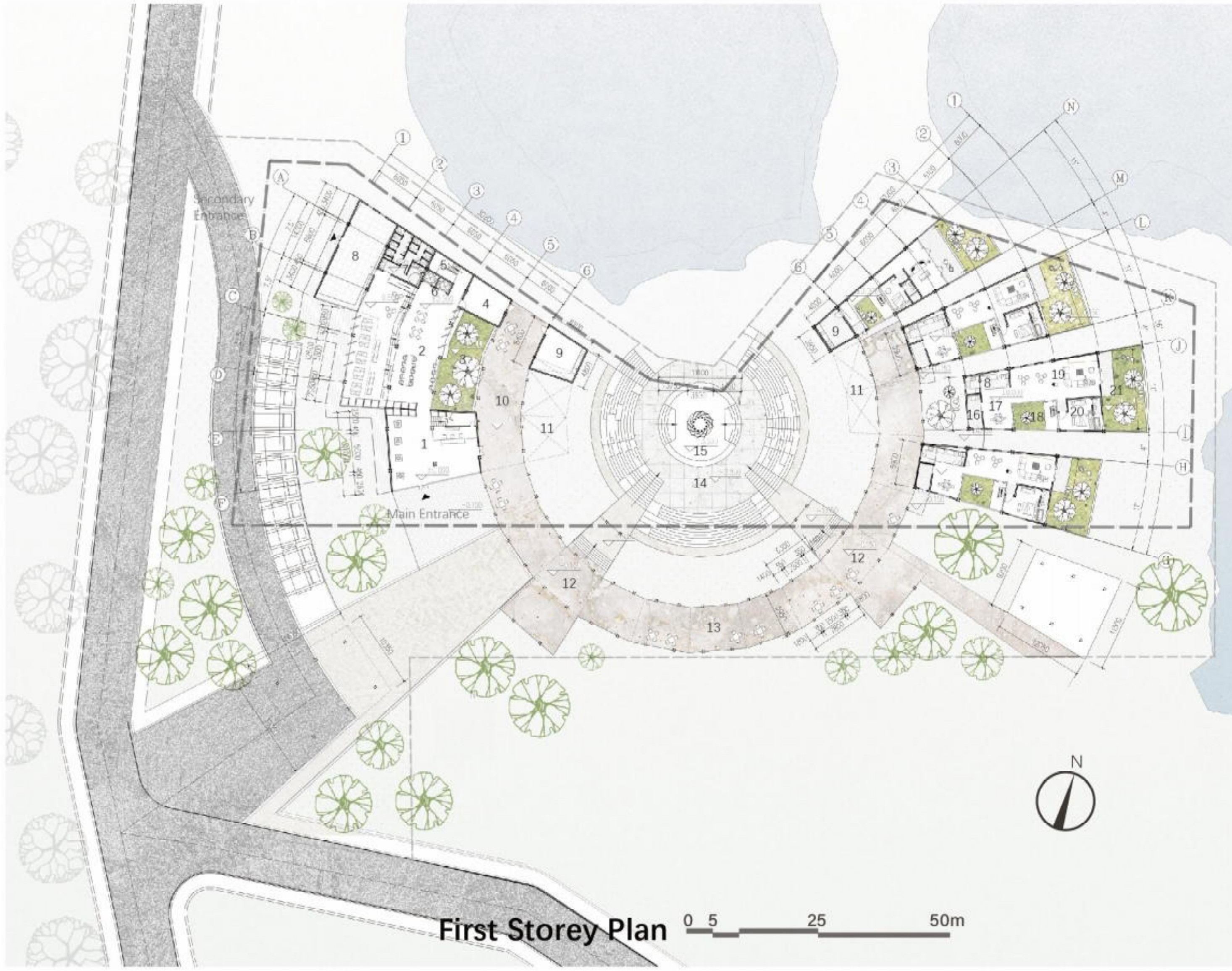
Reversed Dyson Sphere

High-Tech Guided
Solar Resilient Rural Design

Design Description

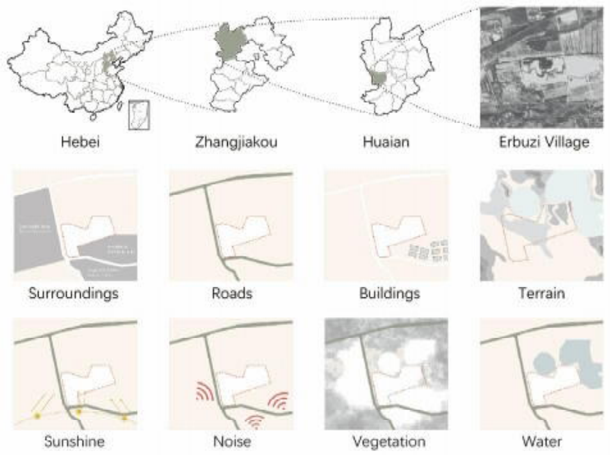
The design is inspired by the Dyson Sphere — a spherical structure intended to capture nearly all the energy emitted by a star — and aims to maximize solar energy capture within a limited site area. The heliostat system concentrates dispersed solar energy onto the apex of a solar tower, where molten salt is heated for power generation, with the energy flow direction being opposite to that of the Dyson Sphere. In terms of spatial layout, the public building and guesthouse are arranged radially around the solar tower, situated on opposite sides of the site, and connected by a circular corridor. The design integrates multiple advanced technologies, including molten salt tower solar thermal power generation, seismic-resistant hierarchical design, variable emergency escape pods, and the geothermal heat pump system. These high-tech strategies not only guide the spatial organization and configuration, but also enhance the building's self-sustaining capabilities under extreme environmental conditions, ensuring the resilience and sustainable development of the village.

Site Plan



First Storey Plan

Site Analysis



Industrial Culture

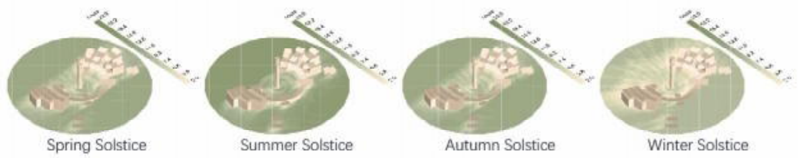
Woodworking Culture

Erbuzi village has a long history of woodworking with a huge local market for woodworking experiential activities.

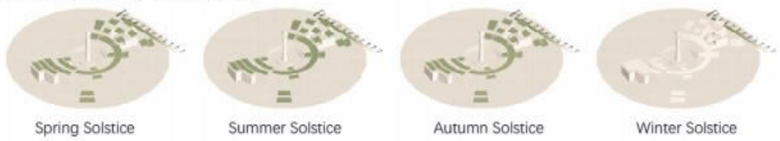
Almond Cultivation Culture

The Site has a rich planting Culture, with the main variety being the flat almonds.

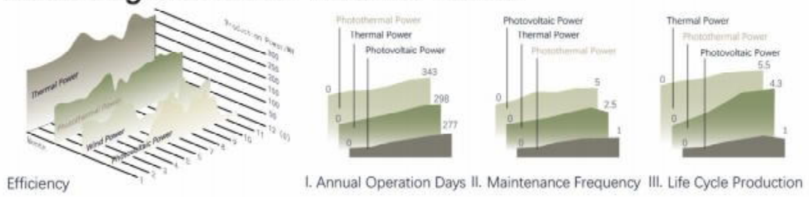
Sun Hours



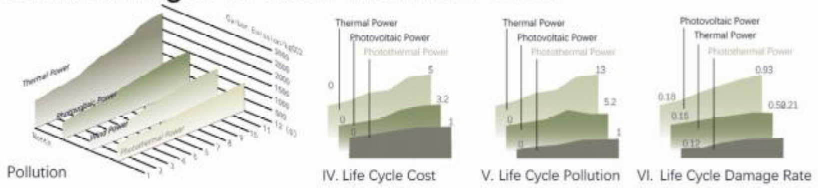
Incident Radiation



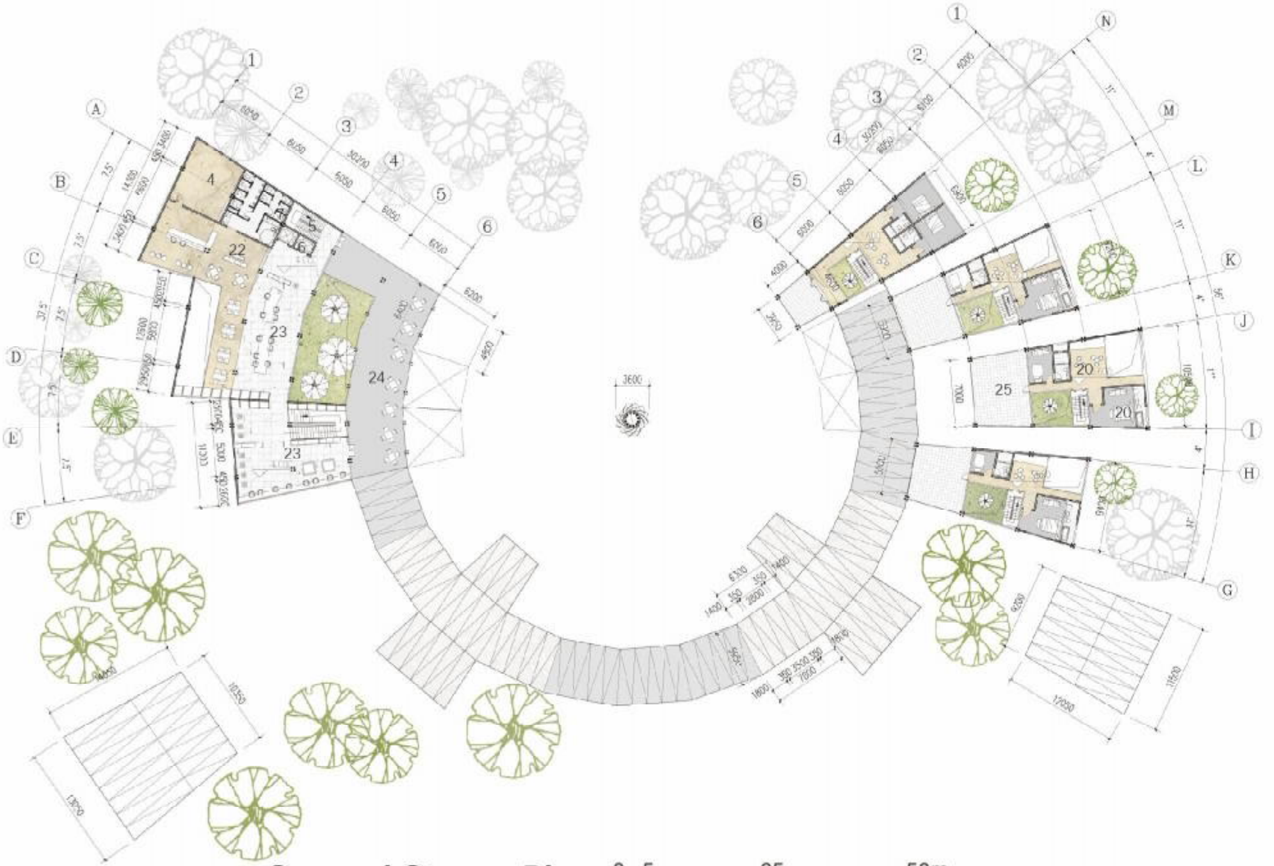
Advantages of Solar Thermal Power



Disadvantages of Solar Thermal Power



1 Entrance Hall
2 Canteen
3 Landscape Courtyard
4 Warehouse
5 Staircase
6 Lifts
7 Restroom
8 Kitchen
9 Medical Station
10 Activity Space
11 Expansion Space
12 Emergency Evacuation Capsule
13 Ring Corridor
14 Square
15 Stage
16 BnB Entrance Hall
17 Dining Room
18 Landscape Sunroom
19 Living Room
20 Bedroom
21 Backyard
22 Coffee Bar
23 Woodworking Experience Space
24 Science Popularization Space
25 Apricot Drying Space

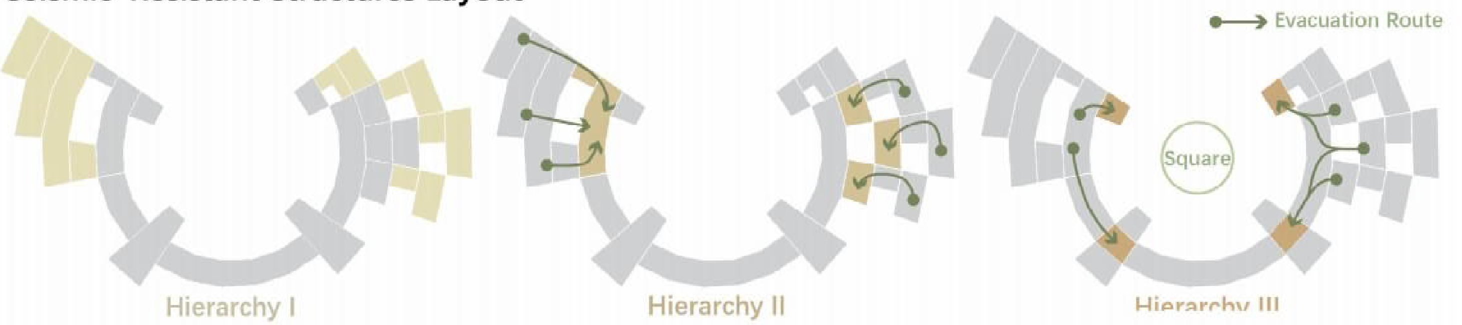


Second Storey Plan

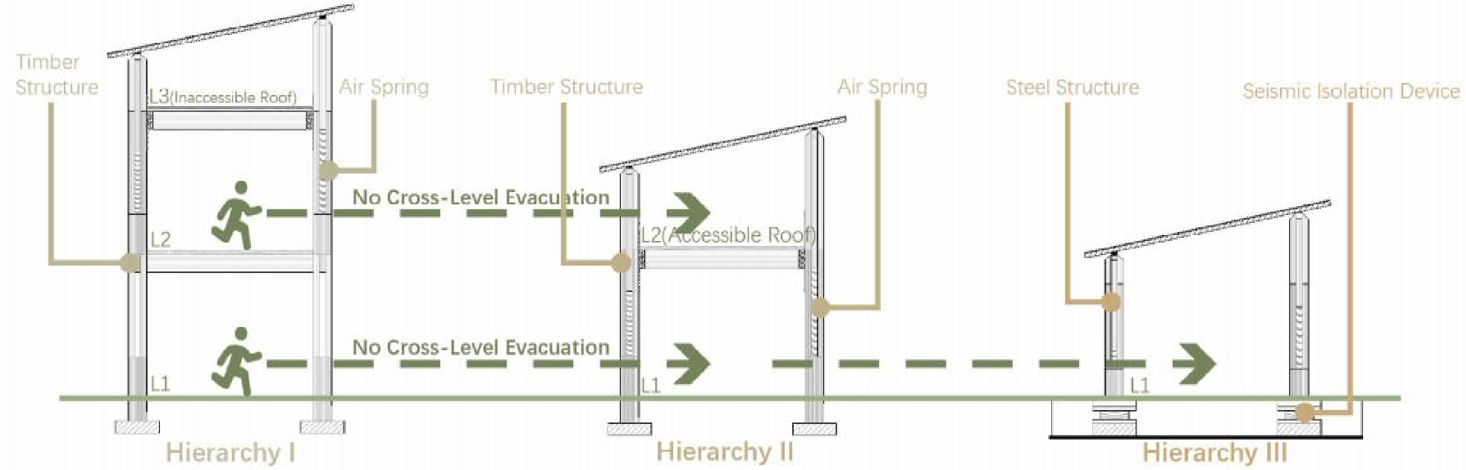

High-Tech Overview
Central Tower
A multifunctional tower that integrates solar thermal power generation, seismic warning and serves as a landmark.
Landscape Sunroom
Make full use of the south-facing sunlit space as a sunroom, ensuring interior space stays warm in winter and cool in summer.
Subdivision of Triangular-Mesh as an Architectural Module
Landscape Courtyard and Double-Height Space
Utilizes passive ventilation by harnssing the movement of warm and coool air.
Apricot Drying Space
Seismic and Hail Resistant Joint
Expandable Medical Station
Carpentry Experience Space
Insulated Water Tank
Condenser
Evaporator
Intake Well
Return Well
Molten Salt Solar Thermal Power Station
Uses heliostats to focus sunlight to heat molten salt at the top of a tower.
Rain Collection System
Make full use of the Site's elevation differences by directing rainwater to collect at the lowest point, where it permeates through permeable pavers into a water tank.
Groud Source Heat Pump
Utilize the constant temperature characteristics of the underground environment, exchanging heat through buried heat exchange pipes to achieve efficient heating and cooling.
Groud Source Heat Pump

Hierarchical Design of Seismic-Resistant Structures

Seismic-Resistant Structures Layout



Seismic-Resistant Structural Analysis



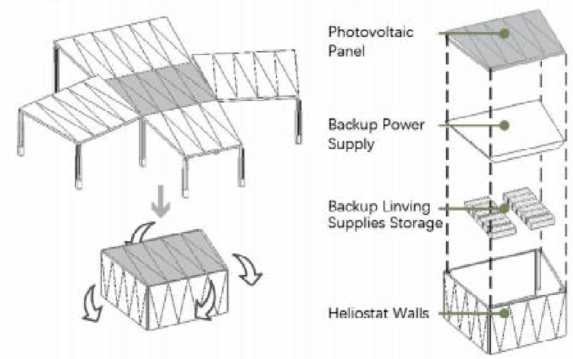
Comparison

	Hierarchy I	Hierarchy II	Hierarchy III
Number of Structural Levels	3	2	1
Structural Self-Weight	Heavy	Medium	Light
Structural Material	Timber	Timber	Steel
Distance from Square	Long	Medium	Short
Special Seismic Design	Air Spring, Timber Joint	Air Spring, Timber Joint	Seismic Isolation Device

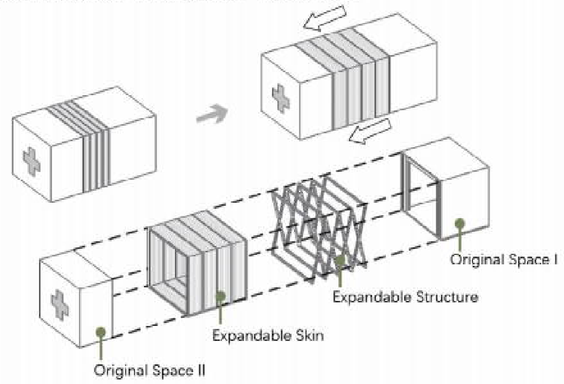
Hierarchical Design of Seismic-Resistant Structures utilizes structural weight, material selectio .etc to achieve cost-effective and efficient evacuation flow organization, which enhances building safety by ensuring quick and safe evacuation through optimized pathways. Additionally, it maximizes resource adaptability, allowing adjustments for varying needs.

Variable Space

Emergency Evacuation Capsule

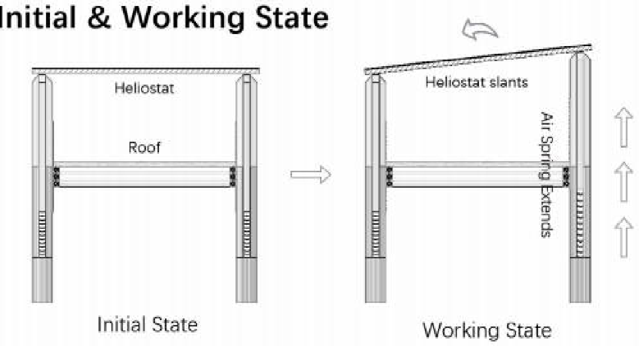


Expandable Medical Station



Seismic and Hail Resistant Joint

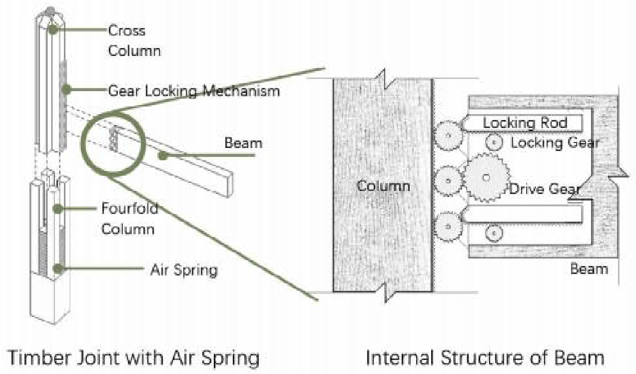
Initial & Working State



Initial State

Working State

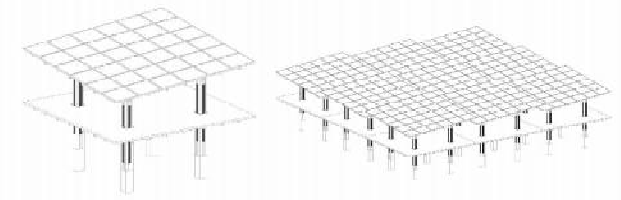
Timber Joint



Timber Joint with Air Spring

Internal Structure of Beam

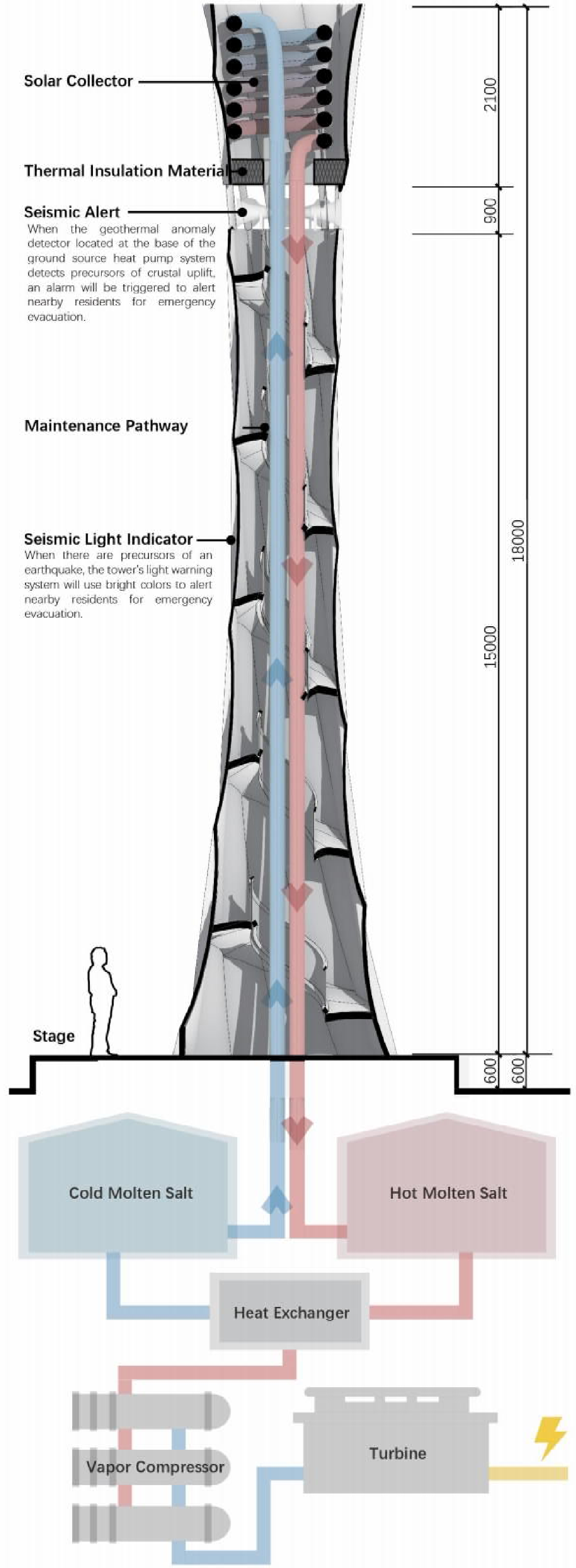
Unit Ensemble



Architertural Unit

Architectural Ensemble

Central Tower



03

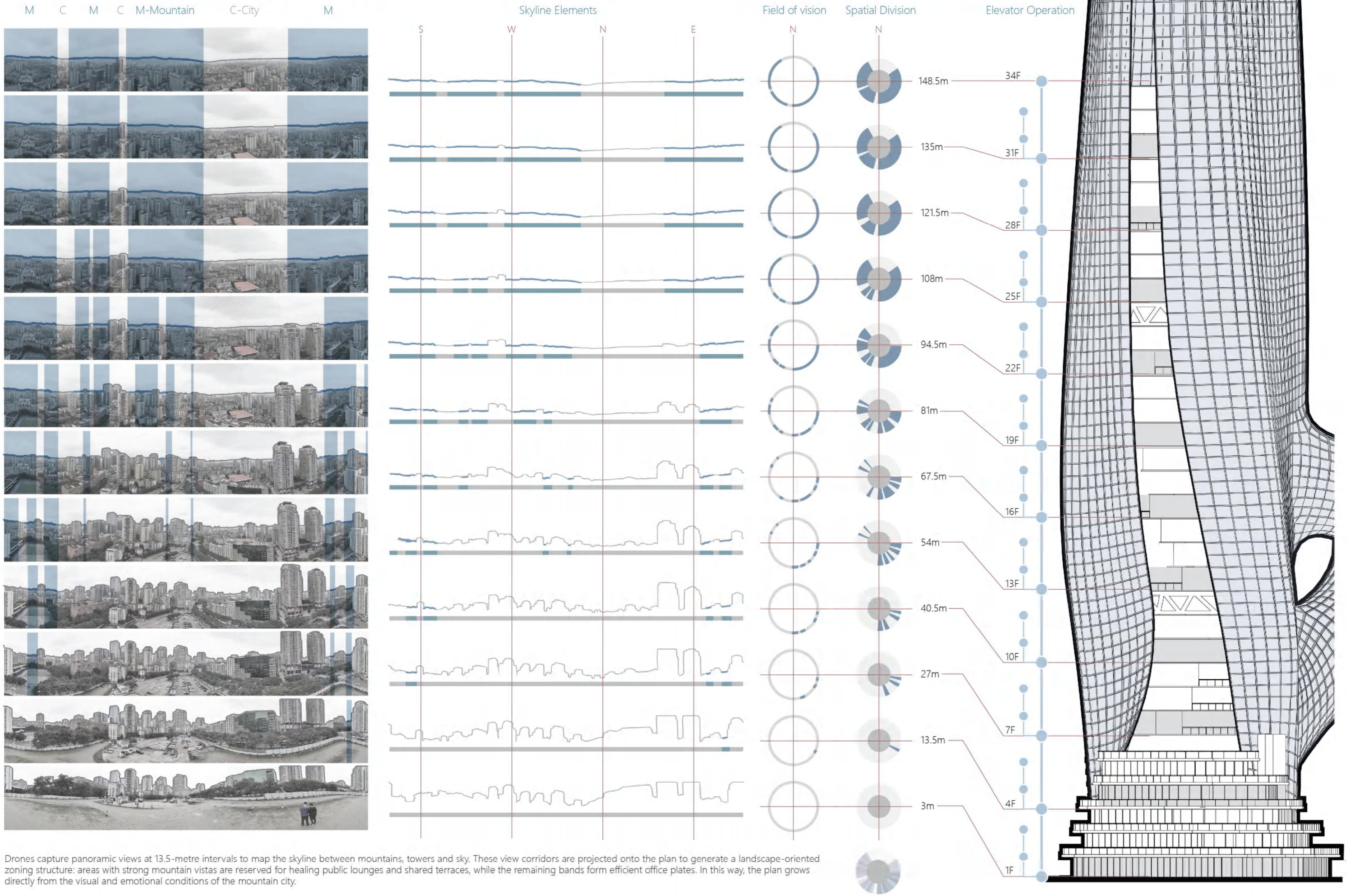
SERENE MOUNTAINS

Design of Super-Tall Buildings
Based on Mountain Skyline Analysis

Design Description

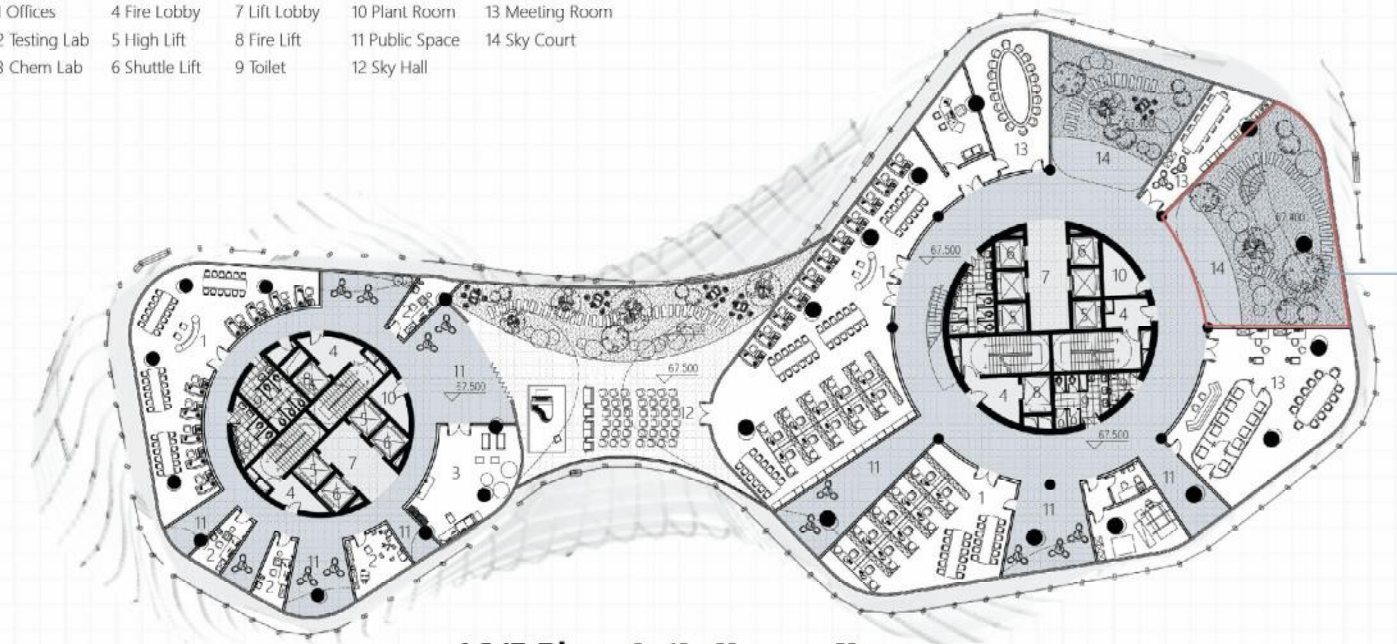
The project aims to create an urban healing centre, serving as a place where people can rebuild connections, share memories and find emotional support. Against this backdrop, the design begins by preserving the mountainous city's cultural context. Using drones to capture panoramic views at varying elevations (each level 13.5m high), it analyses the urban skyline to delineate boundaries between "sky and architecture" and "sky and mountains". This analysis informs the spatial division of the building plan. Sections offering significant landscape views are designated as public leisure spaces, providing psychological respite and brief relaxation for high-pressure office workers; the remaining areas serve as office spaces. In the sectional design, a 13.5-metre unit (comprising three storeys, each 4.5 metres high, meeting fire compartment requirements) was employed. Through AMEBA algorithmic structural topology optimisation of the floor slabs, irregularly sized openings were generated. This approach not only reduces material consumption but also enhances visual connectivity between floors, creating a more fluid and open spatial experience. This responds to people's profound desire for interaction and the rebuilding of social connections.

Spatial Division Strategy based on Skyline Elements



Drones capture panoramic views at 13.5-metre intervals to map the skyline between mountains, towers and sky. These view corridors are projected onto the plan to generate a landscape-oriented zoning structure: areas with strong mountain vistas are reserved for healing public lounges and shared terraces, while the remaining bands form efficient office plates. In this way, the plan grows directly from the visual and emotional conditions of the mountain city.

1 Offices
2 Testing Lab
3 Chem Lab
4 Fire Lobby
5 High Lift
6 Shuttle Lift
7 Lift Lobby
8 Fire Lift
9 Toilet
10 Plant Room
11 Public Space
12 Sky Hall
13 Meeting Room
14 Sky Court



16/F Plan 0 10 20 50m

1 Offices
2 Testing Lab
3 Chem Lab
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7 Lift Lobby
8 Fire Lift
9 Toilet
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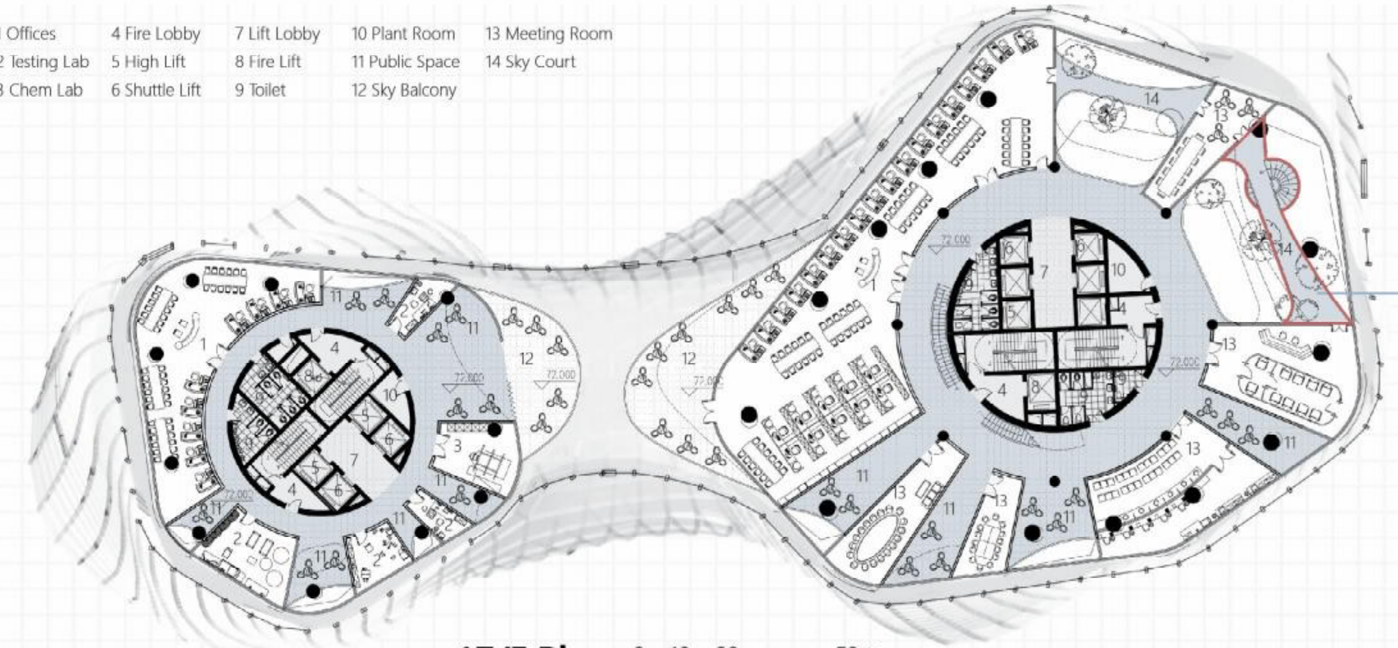
12 Sky Balcony
13 Meeting Room
14 Sky Court



17/F Plan 0 10 20 50m

1 Offices
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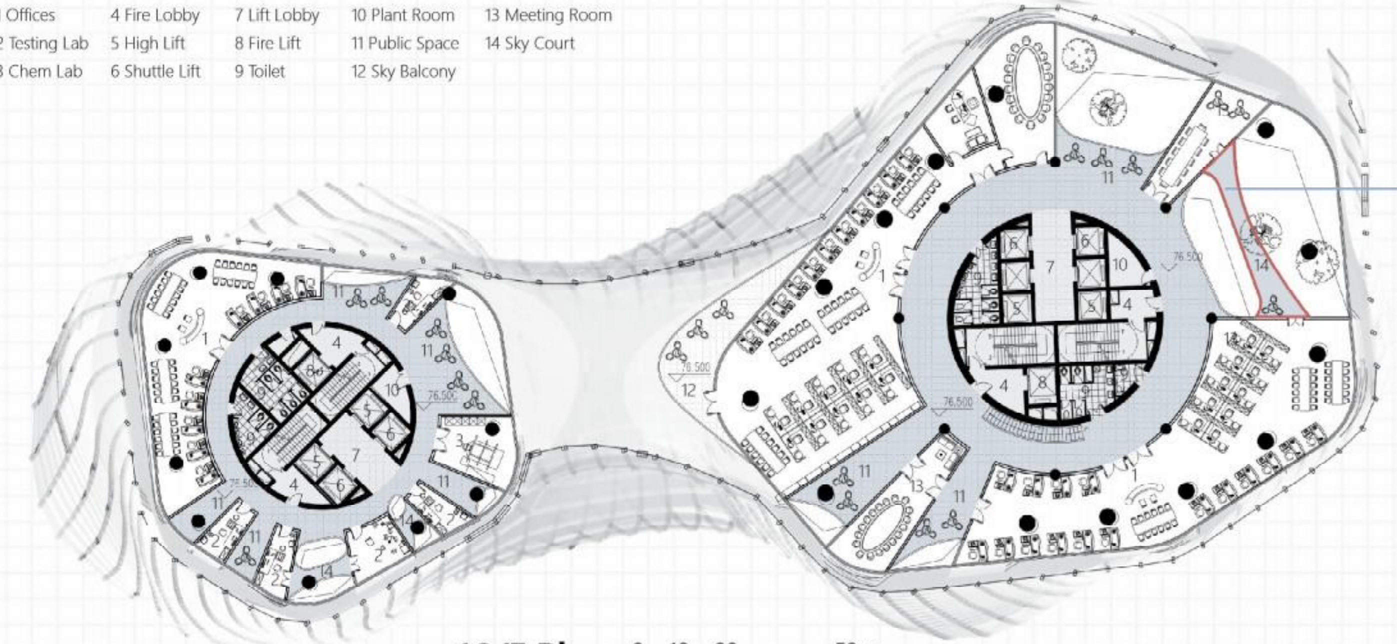
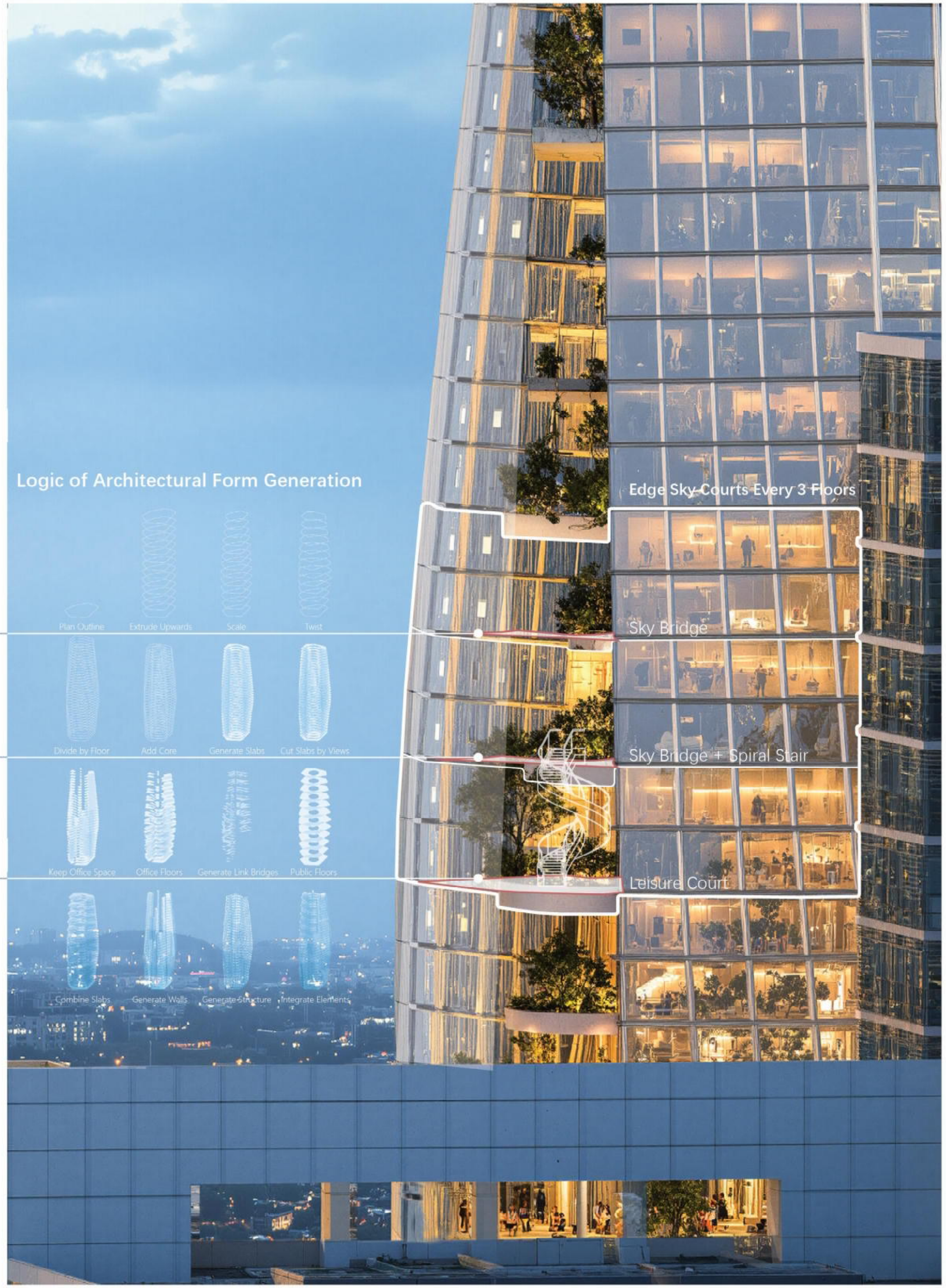
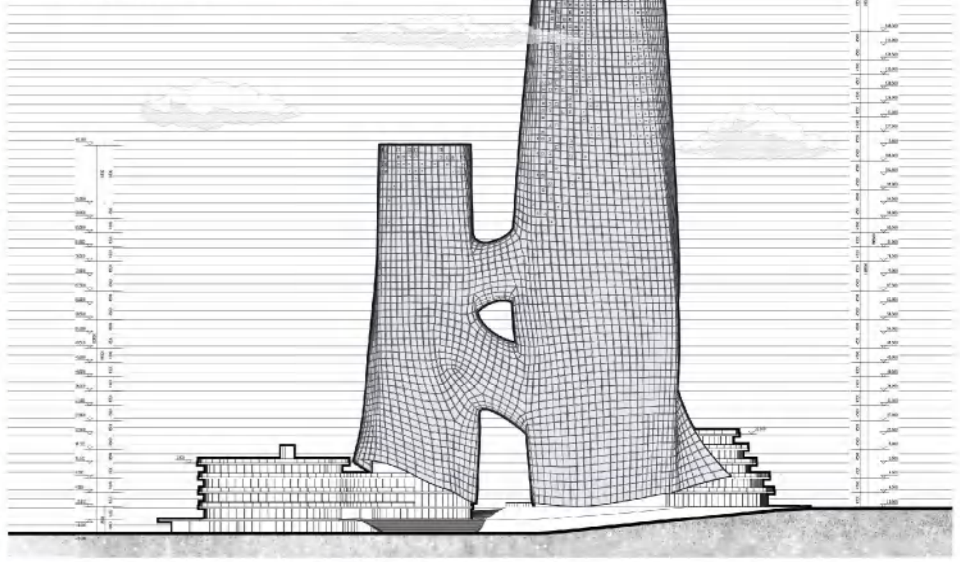


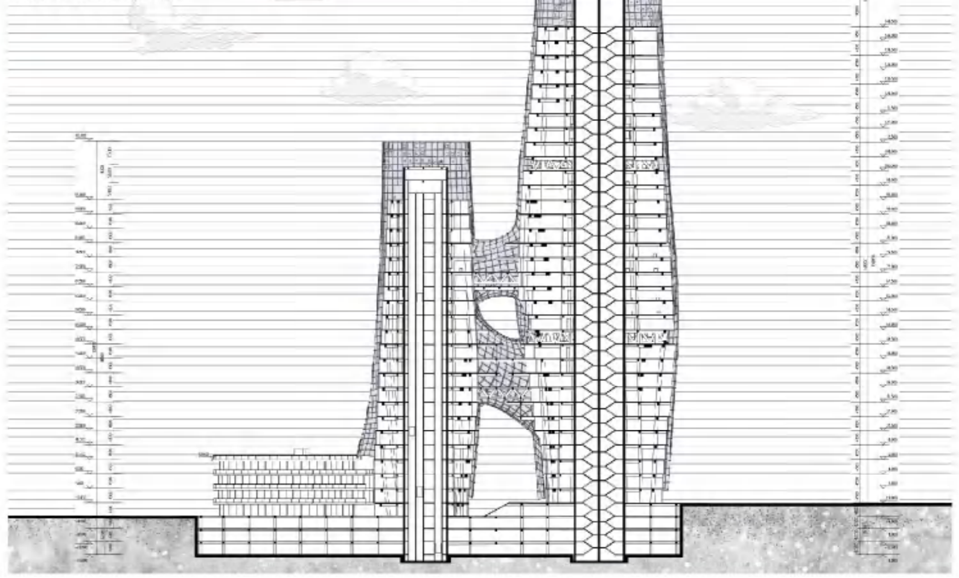
18/F Plan 0 10 20 50m



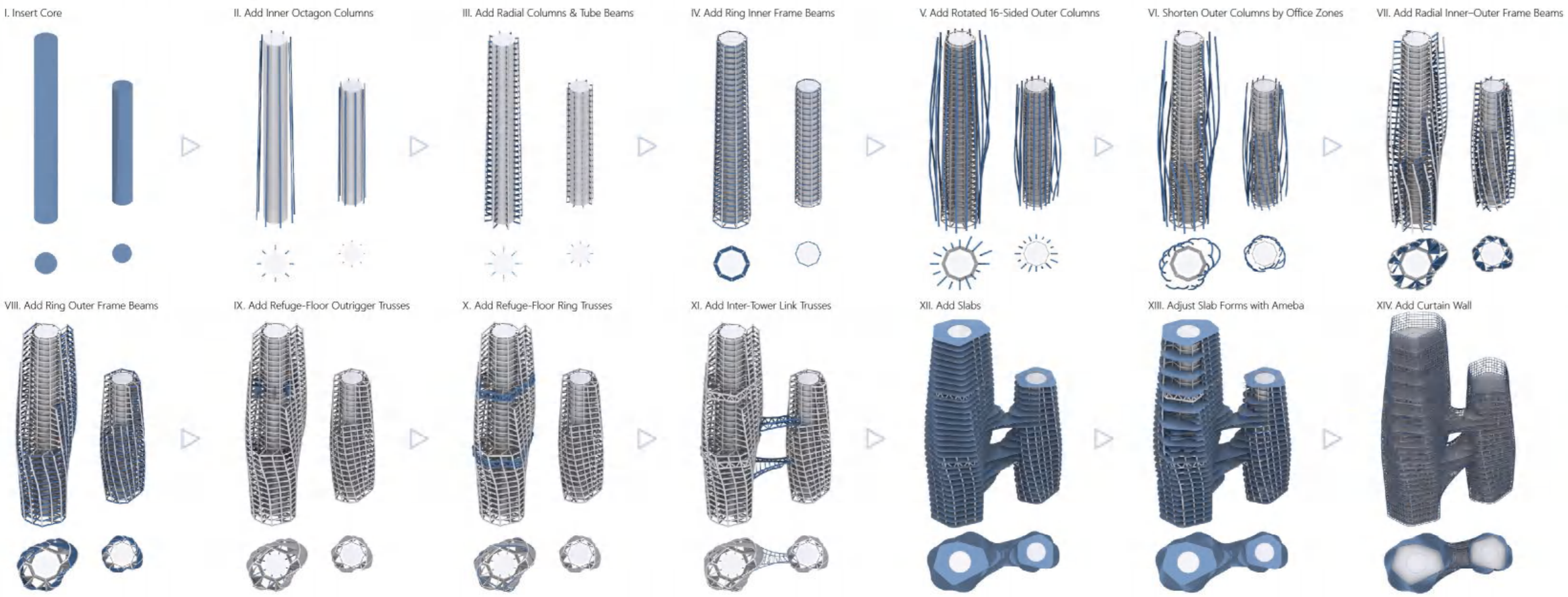
Main Elevation



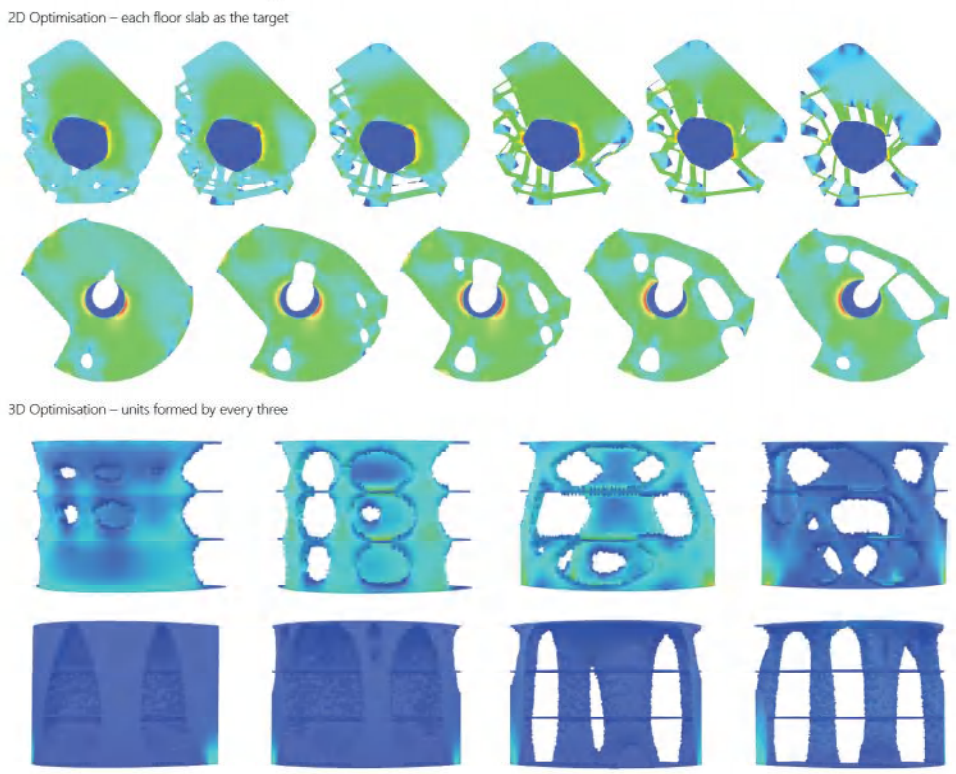
Core Section



Structure Generation Sequence



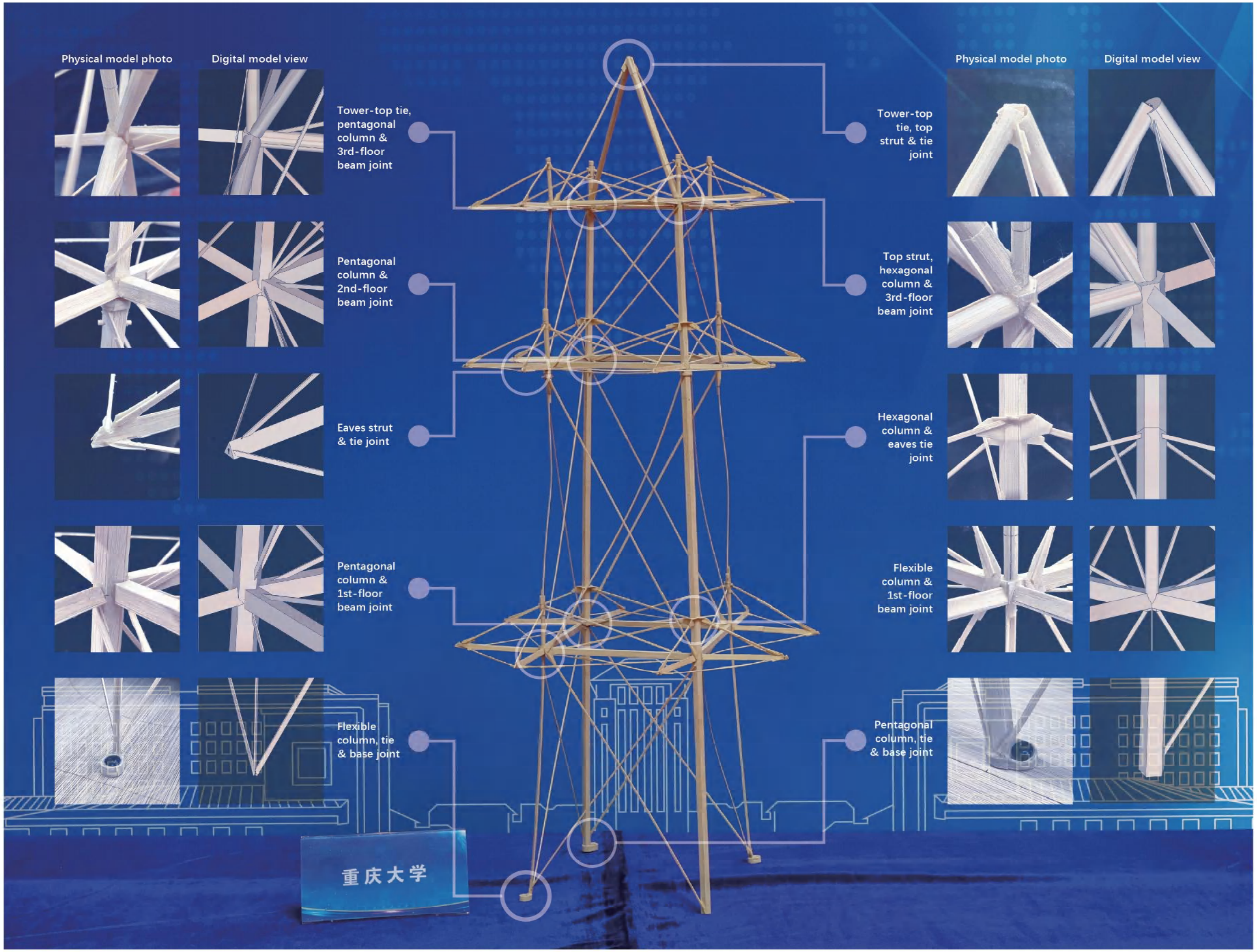
Ameba Structural Optimisation



04

TRIPLE TIMBER TOWER

Spatial Timber Structure under Multiple Load Cases



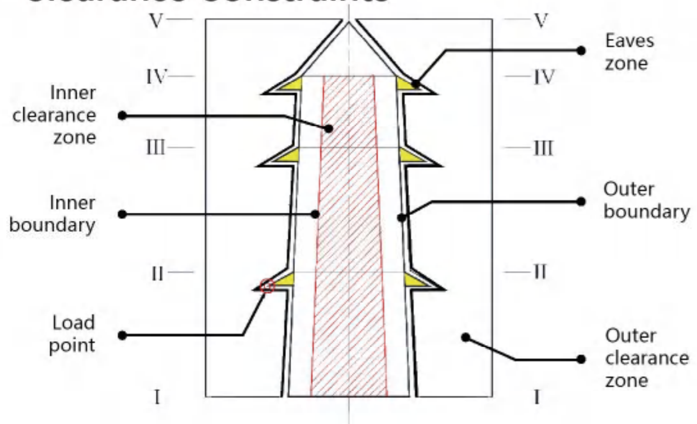
Design Description

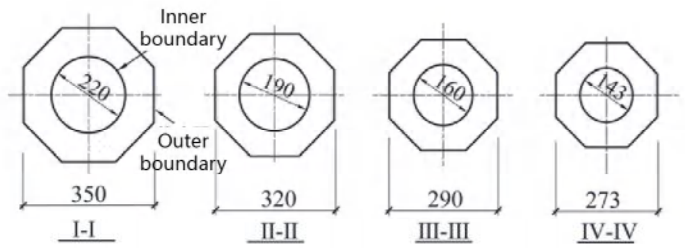
This model is the runaway **first-prize** winning entry in the **20th Chongqing University Structural Design Competition**, weighing only 151.6 g yet safely carrying over 100 kg of various applied loads during testing. Several schemes with four, three, two and single main columns were studied, and a two-column system was finally adopted after comparing stiffness, stability and constructability. The final scheme consists of two rigid main columns, three rings of folded circumferential beams and a tower head, with continuous tension ties placed at the kinks of the folded beams to resist out-of-plane overturning moments and to provide global restraint to the tower.

The two tapered rigid columns are fabricated as hexagonal and pentagonal tubes whose wall thickness gradually decreases with height, reducing self-weight while maintaining adequate strength and stiffness. The first and second rings of circumferential beams use rectangular composite tubes made from bamboo strips and bamboo skin, while the third ring adopts circular bamboo-skin tubes; the cantilevered eaves are braced by triangular tubular struts to improve structural efficiency. All joints are designed as prefabricated connections: sleeves with dowel pins enable rapid assembly of the main joints, and the eave tension ties are anchored to the columns by outer ring plates and slotted insertion details. This ensures clear force paths, ease of fabrication and high on-site assembly accuracy.

Problem Interpretation

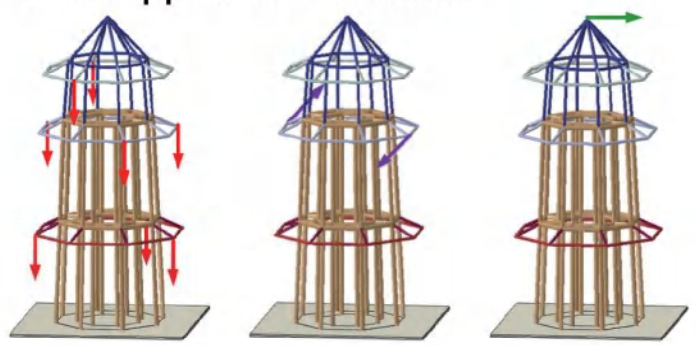
Clearance Constraints





The brief required a spatial timber tower model whose members were placed only between the inner circular clearance and the outer octagonal boundary, with the outer dimensions tapering from bottom to top.

Load Application Methods



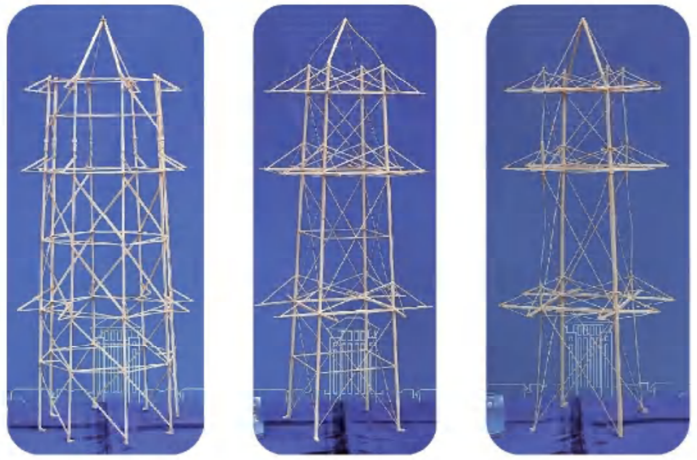
Stage 1 — Vertical Loading
Simulates floor and roof live loads. Three eave points on Levels 1 and 2 and two on Level 3 are randomly selected, each carrying 4 kg.

Stage 2 — Torsional Loading
Simulates torsional wind effects. Opposing 3 kg horizontal forces are applied to two opposite eave points on Level 2.

Stage 3 — Horizontal Loading
Simulates lateral wind thrust. A fixed-direction 7 kg horizontal load is applied at the top.

Design Development

Structural Scheme Options

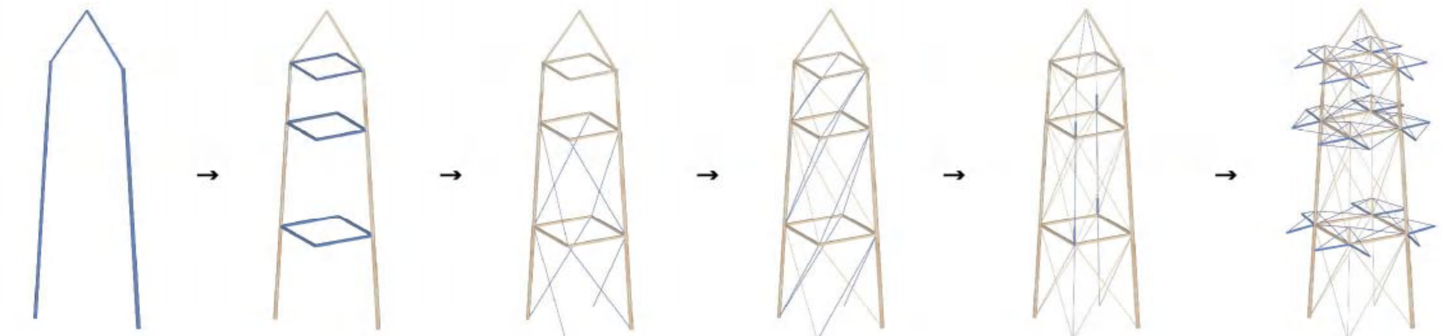


8-Column Scheme
Pros: High inherent stability and redundancy
Cons: Material utilisation is low when eave points are unloaded; large bending inertia requires heavier members

4-Column Scheme
Pros: Stable; eaves struts and beams align well for smooth axial force transfer
Cons: Material efficiency varies strongly with load cases; some members cannot fully mobilise strength

2-Column Scheme
Pros: High material utilisation; strength mobilised
Cons: Lateral stability relies on ties and flexible columns; overall stiffness is lower and deformation more likely

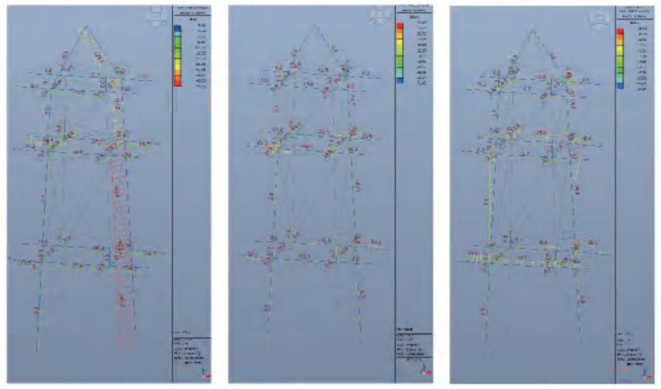
Structural Member Breakdown



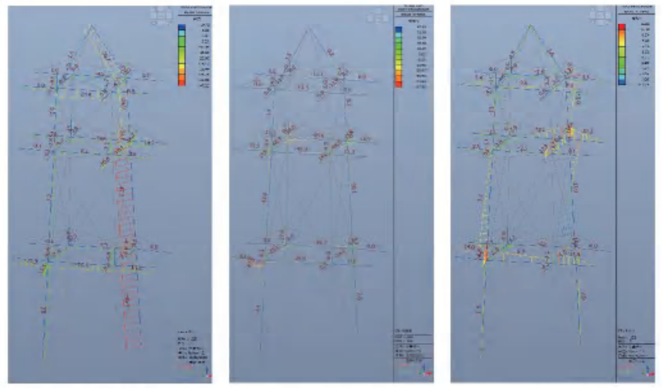
Vertical load path formed by two columns and top → Three ring beams added for shear transfer → Clockwise torsional ties added → Third-direction bending ties added → Flexible columns added for anti-overturning → Eaves struts & balancing ties added

Force Analysis (MIDAS GEN)

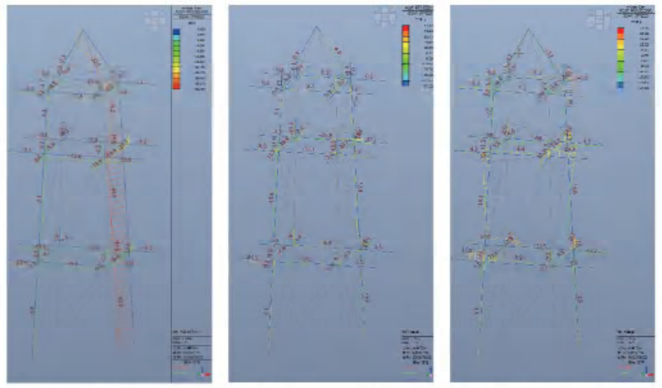
Strength Analysis (30 cases from 300,000, 4 shown)



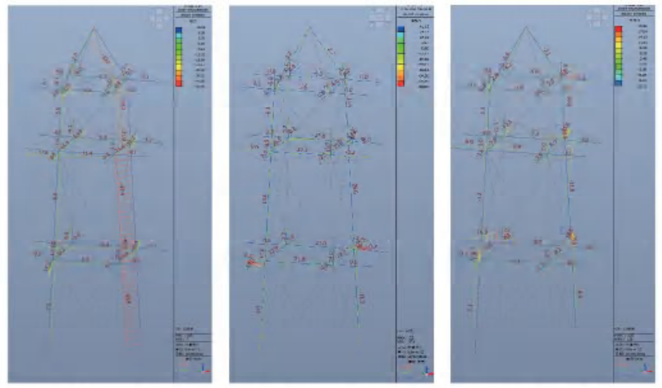
Load Case 1 – Beam Stress (Axial, My, Mz) / MPa



Load Case 2 – Beam Stress (Axial, My, Mz) / MPa



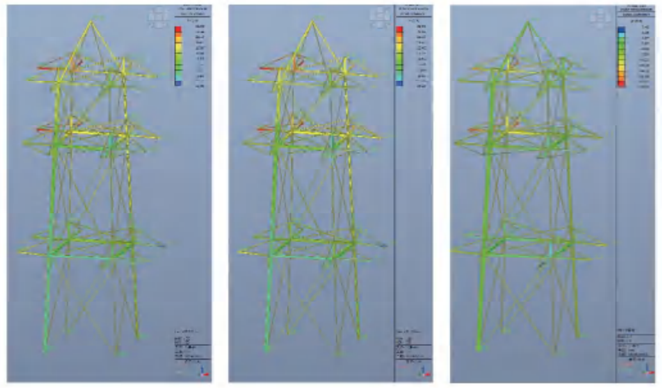
Load Case 3 – Beam Stress (Axial, My, Mz) / MPa



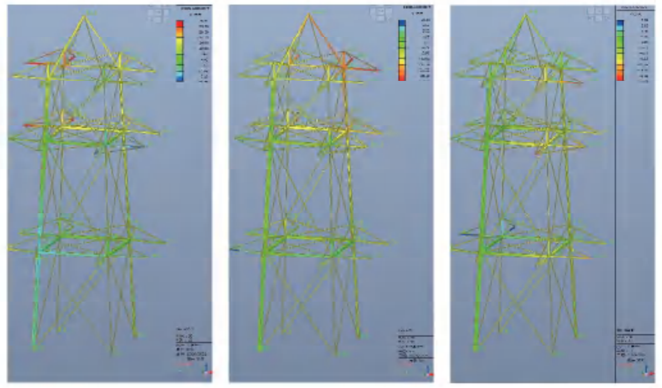
Load Case 4 – Beam Stress (Axial, My, Mz) / MPa

Level 2 beams (1-4 and 3-6) must be the strongest section due to maximum torsional stress from Level 2 loading, while tower-top stiffness is essential for Level 1 load transmission. Bamboo powder filling must be optimized based on whether the component is subject to axial tension or compression.

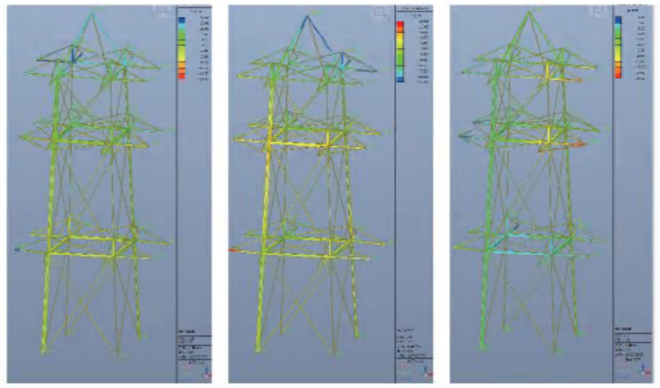
Stiffness Analysis (30 cases from 300,000, 4 shown)



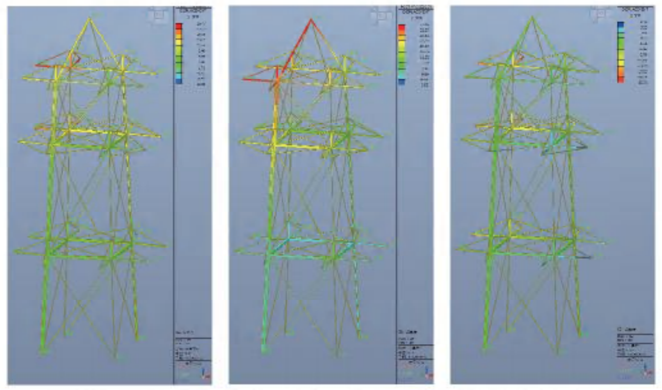
Load Case 1 – Nodal Displacement (DX, DY, DZ) / mm



Load Case 2 – Nodal Displacement (DX, DY, DZ) / mm



Load Case 3 – Nodal Displacement (DX, DY, DZ) / mm



Load Case 4 – Nodal Displacement (DX, DY, DZ) / mm

Eave stays need a high-bending-stiffness cross-section (e.g., triangular tube). The tower top must possess sufficient rigidity to prevent overturning, especially when column loads are concentrated. Furthermore, all structural components must ensure adequate strength at connections, where the specific bamboo powder filling technique must be optimized for axial stress: filling side edges for tension and the contact surface for compression.